

Casting Defect Detection using a Fine-Tuned Vision-Language Model

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1. AIM

To measure **how well a pretrained vision-language model detects casting defects before any training, and how much task-specific fine-tuning changes that** — with the base model measured first, and the fine-tuned model evaluated on exactly the same images with exactly the same prompts, so the difference is attributable to fine-tuning alone.

2. WHAT WAS USED

Model	Qwen/Qwen3-VL-4B-Instruct — 4B params, Apache-2.0, ungated
Why this one	Ungated + permissive licence, current documented API, fits a free Colab T4
Dataset	Kaggle <i>simmshaikh/casting-defect-detection</i> , CC BY-SA 4.0
Libraries	PyTorch, Transformers 5.x, PEFT (LoRA), scikit-learn, Gradio
Hardware	Analysis on MacBook Air M4 (CPU); inference & training on Colab NVIDIA T4
Method	LoRA / parameter-efficient fine-tuning, vision tower frozen

3. DATASET — MEASURED, NOT ASSUMED

Labelled images	1,200 (the card's "1,493" also counts unlabelled overlays and plots)
Classes	12 — ok + 11 defect types, exactly 100 each
Image size	512×512 RGB JPEG, uniform; 0 corrupted
Binary balance	100 OK vs 1,100 Defective (1:11)
Split	train 838 / val 184 / test 178, seed 42

4. MECHANISM

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UPLOAD CASTING IMAGE → VISION-LANGUAGE MODEL → parse structured reply →  
  TRACK A (primary): Is the casting defective? → OK / Defective  
  TRACK B (secondary): which of the 12 defect types? → label or "Unmatched"  
  → model-reported confidence + visual evidence
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The model receives one image plus one of three fixed prompts and replies in a structured format, parsed into a verdict, defect type, confidence and justification. Fine-tuning teaches that verdict from the dataset's own ground-truth labels — no label is invented.

5. WHAT WAS DONE

- Dataset audit.** Every count and dimension measured from the files themselves; verified byte-identical to the authentic Kaggle archive by checksum.
- Leakage prevention.** Filenames carry *no* source-casting identity (index-match rate 2.9% = chance), yet 360 cross-class pairs exceed 0.99 similarity — the same casting reused across classes. Images are therefore grouped by *visual content* into 854 groups, each assigned wholly to one split. Verified: worst cross-split similarity 0.9491, zero groups spanning splits.
- Frozen evaluation.** 178 held-out test images × 3 fixed prompts = **534 generations**, locked to disk so the base and fine-tuned runs see identical inputs. Greedy decoding, seed 42. Nine run settings plus the full decoding config are asserted equal across the two runs before any comparison is accepted.
- Two-track scoring.** Track A on a class-balanced 12 OK + 12 Defective slice; Track B on all 178 images across the 12 original labels.
- Engineering.** 166 automated tests; the training script *refuses* to run before a baseline exists; the comparison script *aborts* if the two runs differ in even one image.

6. END RESULT — ALL EXPERIMENTS COMPLETE

DELIVERABLE	STATUS
Dataset analysis + leakage-safe split	Complete
Base-model baseline (534 generations)	Complete
LoRA fine-tuning (315 steps, 73.7 min)	Complete
Fine-tuned evaluation, same images	Complete
Base vs fine-tuned comparison	Complete
Gradio upload-and-predict demo	Built
Generalization on real photographs	Out of scope

Track B — 12-class defect typing: the principal result. The base model sat at chance; the fine-tuned model identifies the synthetic defect type reliably.

12-CLASS DEFECT TYPE	BASE	FINE-TUNED
Accuracy	0.1220	0.8333
Macro F1	0.0371	0.8144
Coverage	0.4607	0.6742

Chance level for 12 classes is 0.083. Coverage = share of replies naming a usable class; prompts never listed the 12 names.

7. TRACK A — THE BINARY RESULT MUST BE READ ON BOTH VIEWS

BINARY OK VS DEFECTIVE	BASE	FINE-TUNED	Δ
Balanced slice (12 OK + 12 Def) — primary view	0.7361	0.5833	−0.1528
Macro F1, balanced slice	0.7361	0.4958	−0.2403
Full test set (166 Def / 12 OK) — secondary	0.6330	0.9176	+0.2846
Recall, Defective	0.6245	0.9719	+0.3474
Recall, OK	0.7500	0.1667	−0.5833
False negatives / false positives	187 / 9	14 / 30	−173 / +21

The two accuracy rows are the same behaviour seen from two angles. The fine-tuned model answers "Defective" almost unconditionally: it catches every defect in the balanced slice and recognises only 6 of 36 OK rows. A test set that is 93% Defective rewards that; a balanced one does not. **The +0.2846 figure must not be quoted without the −0.1528 beside it.** The cause is identifiable: the training split is 767 Defective vs 71 OK (91.5% Defective), and no class-balanced sampling or loss weighting was applied. All 24 regressions are OK castings, and none are Defective — the signature of a learned base rate rather than a learned feature.

8. WHAT FINE-TUNING CHANGED, PER IMAGE-PROMPT PAIR (534 ROWS)

QUADRANT	ROWS	COMPOSITION	ALSO MEASURED	BASE	FINE-TUNED
Both correct	314	—	Accuracy spread across 3 prompts	0.3596	0.0225
Fixed by fine-tuning	176	173 Defective, 3 OK	Mean response length (chars)	372	97
Broken by fine-tuning	24	24 OK, 0 Defective	Generation rate (img-prompt/s)	0.20	0.56
Both wrong	20	mostly f l a s h, the subtlest texture	Confidently-wrong error rows	95	16
			Hallucinated defect type	1	16
			Unparseable outputs	0	0

Prompt sensitivity was largely eliminated — a clean, separable gain. The base model's dominant error was *missing* defects; the fine-tuned model's is *inventing* them, sometimes with a fabricated defect type and bounding box on a defect-free part.

9. HONEST LIMITATIONS

The defect classes in this dataset are synthetic — generated by a script that paints defect textures onto real defect-free castings. Results therefore describe detection of *synthetic* defect patterns, and do not demonstrate recognition of real industrial casting defects. Validation on genuine defect photographs remains future work.

- Only 12 OK images reach the test set, so OK-class recall carries wide error bars (Wilson 95% CI on the base model's OK recall: 0.468–0.911) and is reported as indicative.
- The 91.5% Defective training split was not corrected for; class-balanced sampling, loss weighting or a calibrated decision threshold are the obvious next steps.
- Confidence values are the model's own self-reported numbers, not calibrated probabilities — one wrong answer was stated at 100% confidence.
- Single random seed; no confidence intervals across repeated runs.
- A conventional CNN would likely be stronger and cheaper here; the VLM is studied because the task is a before/after fine-tuning comparison, not a search for peak accuracy.
- Research prototype only — not a certified industrial quality-control system.

All figures measured from the actual runs and copied from files under results/ — none written by hand. Baseline 2026-08-28 (43.6 min) and fine-tuned evaluation 2026-08-29 (20.5 min), both on a Tesla T4, 534/534 generations each, zero generation errors, zero unparseable outputs. Reproducible pipeline, fixed seed 42, frozen prompts and evaluation subset.